

Surface Area, Composite Shapes, and Scale Design

Answer Key

Grades 6–7 Mathematics

Quick Answers

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|--|--|
| 1. 52 cm^2 | 7. 50 m^2 |
| 2. 150 in^2 | 8. 15 cm |
| 3. 14 m | 9. 63 cm^2 |
| 4. 47 m^2 | 10. 9000 cm^2 |
| 5. 150 ft^2 | 11. 52 m^2 |
| 6. (a) paper needed = 108 in^2 , (b) sheets to buy = 2 | 12. (a) model length = 20, (a) model width = 15, (b) model floor area = 300, — |
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What is verified

11 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 6–7 Mathematics

6.G.A – *problems 1, 2, 4, 6, 7, 10, 11*

7.G.A.1 – *problems 3, 5, 8, 9, 12*

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

- 5. *If they got 30: multiplied the drawing area by the scale once instead of scaling each side*
 - 8. *If they got 960: multiplied the real length by the scale instead of dividing*
 - 9. *If they got 21: multiplied the area by the scale factor 3 instead of its square*
 - 10. *If they got 10800: included a glass top, but the aquarium is open*
 - 11. *If they got 64: treated the house shape as the full 8 by 8 bounding rectangle*
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1. Closed box $4 \times 3 \times 2 \text{ cm}$: surface area.

Solution: A closed box has three pairs of matching faces: $SA = 2(\ell w + \ell h + wh) = 2(4 \cdot 3 + 4 \cdot 2 + 3 \cdot 2) = 2(12 + 8 + 6) = 2 \cdot 26$, so 52 cm^2

2. Cube with 5-inch edges: surface area.

Solution: A cube has 6 identical square faces: $SA = 6s^2 = 6 \cdot 5^2 = 6 \cdot 25$, so 150 in^2

3. Blueprint scale 1 cm : 4 m; wall 3.5 cm on paper.

Solution: Each blueprint centimeter stands for 4 real meters: $3.5 \cdot 4$, so the wall is 14 m long.

4. L-shaped patio with corners $(0, 0)$, $(8, 0)$, $(8, 4)$, $(5, 4)$, $(5, 7)$, $(0, 7)$.

Solution: Split at the height $y = 4$: a bottom rectangle $8 \times 4 = 32$ and a top-left rectangle $5 \times 3 = 15$. Total $32 + 15$, so 47 m²

5. Floor plan 1 in : 5 ft; bedroom 2 in by 3 in.

Solution: Scale each side first: real length $2 \cdot 5 = 10$ ft, real width $3 \cdot 5 = 15$ ft. Area $= 10 \cdot 15$, so 150 ft²

(Scaling the drawing area $2 \cdot 3 = 6$ by 5 only once gives 30 — too small, because *both* sides stretch.)

6. Gift box $6 \times 4 \times 3$ in: paper, then sheets.

Solution (a): $SA = 2(6 \cdot 4 + 6 \cdot 3 + 4 \cdot 3) = 2(24 + 18 + 12) = 2 \cdot 54$, so 108 in²

Solution (b): $108 \div 60 = 1.8$ sheets, and partial sheets are not sold, so round *up*: buy 2 sheets.

7. T-shaped stage with corners $(0, 0)$, $(10, 0)$, $(10, 3)$, $(7, 3)$, $(7, 8)$, $(3, 8)$, $(3, 3)$, $(0, 3)$.

Solution: Split at height $y = 3$: bottom bar $10 \cdot 3 = 30$ and upper column $4 \cdot 5 = 20$ (width $7 - 3 = 4$, height $8 - 3 = 5$). Total $30 + 20$, so 50 m²

8. Bridge 120 m real, scale 1 cm : 8 m: drawing length.

Solution: Going from real to drawing means *dividing* by the scale: $8x = 120$, so $x = 120 \div 8$, giving 15 cm

(Multiplying instead gives 960 — a drawing longer than the bridge.)

9. Logo of area 7 cm² enlarged by scale factor 3.

Solution: Lengths triple, so area multiplies by $3^2 = 9$: $7 \cdot 9$, giving 63 cm²

(Multiplying by 3 only once gives 21, which forgets that both dimensions grow.)

10. Open-top aquarium $60 \times 30 \times 40$ cm: glass area.

Solution: Bottom: $60 \cdot 30 = 1800$. Front and back: $2 \cdot 60 \cdot 40 = 4800$. Two ends: $2 \cdot 30 \cdot 40 = 2400$. No top. Total $1800 + 4800 + 2400$, so 9000 cm²

(Adding a top gives 10800 — but this tank is open.)

11. House-shaped sign with corners $(0, 0)$, $(8, 0)$, $(8, 5)$, $(4, 8)$, $(0, 5)$.

Solution: Rectangle below: $8 \cdot 5 = 40$. Triangle roof: base 8, height $8 - 5 = 3$, area $\frac{1}{2} \cdot 8 \cdot 3 = 12$. Total $40 + 12$, so 52 m²

12. Challenge: 1 : 20 model of a 4 m by 3 m room.

Solution (a): Convert, then divide by the scale: 4 m = 400 cm, and $400 \div 20$ gives

20

cm; 3 m = 300 cm, and $300 \div 20$ gives

15

cm.

Solution (b): Model floor area = $20 \cdot 15$, so

300

 cm^2 .

Solution (c): (*instructor-judged*) Both the length and the width are divided by 20, so the area is divided by $20 \cdot 20 = 400$, not by 20. Any explanation naming the squared effect of a scale factor on area earns full credit.